

“Machine Learning approaches for accelerating antibiotic susceptibility testing (AST) in microfluidic systems”

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Abstract. This work project applies different machine learning approaches to microfluidic bacterial growth data to enable early prediction of minimum inhibitory concentration (MIC), reducing the time required for antibiotic susceptibility testing (AST).

Keywords: Antibiotic susceptibility testing (AST); Bacterial growth; Microfluidic systems; time-lapse imaging; Machine Learning; Random Forest;

1 Introduction and state-of-the-art

Antibiotics revolutionised medicine, but their misuse has accelerated antimicrobial resistance, contributing to 4.95 million associated deaths in 2019. Effective treatment depends on antibiotic susceptibility testing AST, yet phenotypic methods based on broth microdilution require 36–72 hours to determine the MIC, forcing clinicians to use broad-spectrum antibiotics and aggravating resistance. Faster AST is therefore urgently needed. [1]–[3]

Phenotypic AST takes between 16-24h and consists of quantifying bacterial growth in the presence of antibiotics using the gold standard broth microdilution method to determine the minimum inhibitory concentration (MIC). [4] New approaches aim to overcome culture-based limitations by enabling faster, real-time bacterial monitoring, using electrochemical biosensors, nanotechnology platforms, advanced imaging, and microfluidic systems with optical or electrochemical results. Namely, *Li et al. (2025)* developed a real-time digital microfluidics system that completes an AST in nearly 4h, while *Meng et al. (2026)* developed a fully automated system using time-lapse imaging of microcolony growth morphology to obtain MIC within 3h. [5]–[11]

In this context, the microcapillary film (MCF)-based AST is an emerging microfluidic approach for rapid, real-time and label-free analysis of bacterial growth. The MCF consists of antibiotic-preloaded microcapillaries arranged in 10 strips with gradient antibiotic concentrations and a control without antibiotic, essential to microcapillary antibiotic susceptibility testing (mcAST) using a Lab-on-a-stick device capable of measuring the MIC. Recent unpublished data show that dip-stick strips immersed in bacterial suspensions enable microscale growth monitoring via dark-field light scattering and

time-lapse imaging. While microfluidic AST enables real-time kinetic analysis and MIC determination, it is not yet standardised by EUCAST and existing approaches still frequently require prior bacterial expansion on agar. Machine learning can complement these phenotypic methods, extracting susceptibility signals from raw outputs to deliver faster, quantitative results. [12]–[17]

Machine learning (ML) is a field within artificial intelligence that allows computers to learn from data, being widely used to analyse biological data, identifying patterns and correlations, supporting data classification and prediction. ML is increasingly used to compress the time-to-result. *Yu et al. (2018)* applied a convolutional neural network to deep-learning video microscopy of single bacterial cells in a nanolitre microfluidic channel, inferring the MIC within 30 minutes rather than the ~12 h required by broth microdilution. *Sturm et al. (2024)* coupled a nano motion sensing platform with a logistic-regression classifier built on automatically selected signal parameters (validated by repeated stratified cross-validation) to return a growth-independent susceptibility call within minutes of computation, at 89.5–98.9% accuracy on independent isolates. Extending phenotypic prediction to biofilms, *Vergauwe et al. (2025)* trained an ordinal-regression model on MALDI-TOF mass spectrometry, isothermal microcalorimetry and multi-excitation Raman data to predict the tobramycin MIC of *Pseudomonas aeruginosa* biofilms, reaching up to 97.83% accuracy. Together, these illustrate ML's role in shortening the path to a quantitative, phenotypic susceptibility result. [18]–[27]

1.1 Project Objectives & Problem Description

Conventional AST determines the MIC, the lowest antibiotic concentration that prevents visible bacterial growth, by incubating bacteria across a range of concentrations and reading the endpoint after a fixed, relatively long period (typically 16–24 h). In the miniaturised, capillary-based MCF diagnostic platform, growth in each capillary is monitored over time, producing a growth-related result, the Grey Value Ratio (GVR). The GVR represents the normalised optical signal of each microcapillary and is calculated as the ratio between the sample absorbance grey value (AbG), measured in microcapillaries containing bacteria and antibiotic, and a reference AbG measured in medium-only microcapillaries, according to the following equation:

$$GVR = \frac{AbG(sample) - AbG(reference)}{AbG(reference)}$$

Each time-lapse image was analysed using ImageJ. The RGB channels were separated, and the grey value of each image was quantified to obtain the AbG signal intensity as seen in Figure 1.

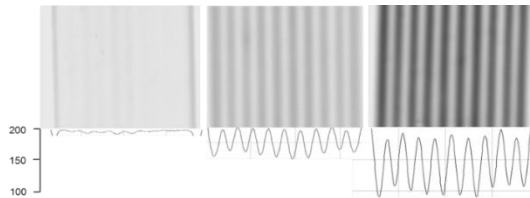


Fig. 1 Split green channel image of MCF, and signal quantification on ImageJ software

The clinical limitation is time. The diagnostically relevant answer (the MIC) is only available at the endpoint, despite the full GVR trajectory being recorded throughout incubation. Inhibited and non-inhibited capillaries often begin to diverge well before the endpoint, raising the main question of this work: can machine-learning models use early points of the GVR time series to predict which capillaries will remain inhibited at the endpoint, and thereby estimate the MIC earlier than conventional reading allows?

A reliable early prediction would shorten time-to-result, which has direct clinical value for initiating targeted antibiotic therapy. Complex as GVR dynamics depend jointly on several interacting factors, species, inoculum, antibiotic, concentration, and time, and early-phase signals are noisier and less separable than endpoint readings. The objective was to organise and analyse experimental GVR data from various bacterial species, concentrations, and antibiotics using Python, and to develop and evaluate machine-learning models capable of predicting per-capillary endpoint inhibition and estimating MIC before the conventional endpoint.

2 Methodology & Computational Tools

2.1 Dataset

This project dataset was obtained during a PhD programme in Chemical Engineering at the University of Bath. It consists of light scattering time-lapse imaging of bacterial growth in the MCF, with measurements every 10 minutes. Two bacterial species were studied: *Escherichia coli* and *Staphylococcus aureus*. *E. coli* was tested against 5 antibiotics, while *S. aureus* was tested against 6. Antibiotics were previously incorporated into the MCF and mixed with bacteria upon loading into microcapillaries.

Bacterial concentrations of 5×10^4 , 5×10^5 , and 5×10^6 CFU/mL were used across experiments. For *E. coli*, ampicillin and tetracycline were each tested in 9 experiments conducted at different concentrations. Ciprofloxacin, gentamicin, and trimethoprim were tested in 3 experiments each at 5×10^5 CFU/mL. For *S. aureus*, ampicillin, ciprofloxacin, gentamicin, and trimethoprim were tested in 5 experiments each across the 3

concentration levels, while tetracycline was tested in 7 experiments and sulfamethoxazole in 3, each at varying concentrations. Overall, there were 27 experiments for *E. coli* and 30 for *S. aureus*, giving a total of 57 experiments. Time series ranged from 11 hours (*E. coli* with ampicillin) to nearly 60 hours (*S. aureus* with ampicillin). The data were available in both raw (AbG sample) and GVR format. Each microcapillary strip consists of 10 microcapillaries, as observed in Figure 2, with a gradient of antibiotic concentration, including a control with 0 $\mu\text{g/mL}$, which is the reference control.

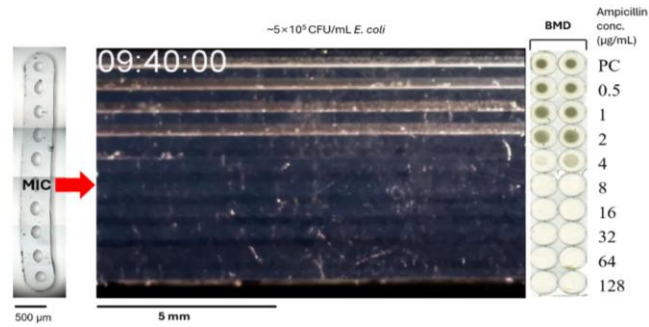


Fig. 2. Time-lapse image (09:40h) of the microcapillary strip loaded with an ampicillin gradient for the 10 microcapillaries and an *E. coli* concentration of 5×10^5 CFU/mL. The yellow zones represent the bacterial growth in the microcapillary. Each microcapillary is compared to the BMD (Broth Microdilution) result with the same bacterial and antibiotic concentration. Font: Zhuoling Yu, University of Bath, 2026

2.2 Data preprocessing

The Excel files were processed in Python and organised into dictionaries indexed by bacteria, antibiotic, date, bacterial concentration, and data type. Each file contains data for a bacterial–antibiotic combination, and each worksheet is split into raw optical measurements (AbG sample) and GVR. Empty rows and columns were removed, and missing values were checked, with none detected. Since GVR is already a normalised form of the raw data, no further transformation was applied. An exploratory data analysis was performed to identify the antibiotics used, experimental dates, bacterial concentrations and experimental durations. Time-series plots of the GVR were generated. The final datasets of *E. coli* and *S. aureus* were saved in two .pkl files.

2.3 Feature Engineering

The ML model will be trained on time-lapse light-scattering imaging data for each microcapillary, using one entry per time point in the experiment. The model will learn how the GVR evolved with individual microcapillaries over time. The features will

capture temporal GVR dynamics associated with bacterial growth/inhibition. This project will build a binary classification ML model that predicts which microcapillaries will grow at the earliest endpoint possible. An incremental development strategy was adopted. The complete methodology was first applied to the *E. coli*-ampicillin dataset to develop and refine the machine learning pipeline. Once established, the same methodology was applied to the full *E. coli* dataset with all five antibiotics. Allowing the model development process to be validated on a smaller dataset before being extended to a larger dataset. The following sections refer to the *E. coli*-ampicillin dataset.

Initial machine-learning development was performed using *E. coli* growth data exposed to different concentrations of ampicillin on the MCF platform. This data corresponds to 9 experiments, with 8420 data points. The initial features included experiment ID, microcapillary ID, time, antibiotic concentration, and GVR. From GVR and time, additional features were derived per microcapillary, including standard deviation, slope (rate of change from the initial GVR), delta (absolute change from the initial value), AUC (area under the curve), and derivative (instantaneous rate of change between time points). Preliminary analysis indicated that positive AUC values were associated with bacterial growth and were therefore used to define the binary labels. Microcapillaries with AUC bigger than 0 were labelled as growth (1), whereas those with AUC less than 0 were labelled as no growth (0). In the *E. coli*-ampicillin dataset, experiment 20230528_5e5 was the only exception and was treated as an outlier. The final dataset contained 8420 rows and 13 columns, with 4949 growth and 3471 no-growth labels. No duplicates or missing values were detected.

2.4 Model Selection

To prepare the *E. coli*-ampicillin dataset for ML, the outlier was removed, and experiment and microcapillary IDs were excluded to avoid data leakage. The data were split by experiment into a training set and two independent test sets (20230518_5e4 and 20230528_5e6), ensuring that evaluation was performed on unknown microcapillaries. The training set contained 5450 rows, while the test sets contained 990 rows each. Classification models were then developed to predict microcapillary growth, including Logistic Regression (max_iter=5000), Random Forest (n_estimators=100, random_state=42), Support Vector Classification (SVC) (RBF kernel, probability enabled), K-nearest neighbours (k=5, with scaled data), and Naïve Bayes. The machine learning was originally tested by all timepoints in the dataset, first with logistic regression and random forest. Feature importance analysis was used to assess potential bias toward antibiotic concentration, which was removed as a training feature due to its impact on model performance. The models were retrained to compare and evaluate classifier performance across the dataset with accuracy, F1-score, recall, precision, AUC-ROC, confusion matrix, and cross-validation scores, with Random Forest achieving the highest performance across all metrics.

2.5 Model Testing

The random forest model was retrained on the same dataset and feature set but evaluated per time point for the 10 microcapillaries. The aim was to identify the earliest time point at which prediction becomes reliable. The training loop was initiated from the first time point where the standard deviation exceeded 0.05, indicating the onset of significant growth-related changes. The model was then evaluated to determine how quickly it could achieve confident predictions, defined by accuracies of 0.90 and 1.00. The overall objective was to predict microcapillary growth or non-growth as early as possible. Initial testing used test sets 1 and 2 with the original features, and plots of accuracy over time were obtained to analyse the results. To improve the results, additional feature engineering was performed.

2.6 Additional Feature Engineering and Testing

To improve predictive performance and achieve earlier predictions, additional features were derived from the initial set. These included time-normalised versions of AUC, delta, and derivative, as well as GVR statistics (maximum, minimum, and mean) computed across microcapillaries and experiments, and a growth score calculated both across and per microcapillary. Several models were then trained using different feature combinations, including the initial features alone, initial features plus all additional features, and subsets of the most informative features (top 15, top 2, `gvr_max_tube`, `auc_per_time`, without `initial_std` and derivative). Accuracy-over-time plots were generated to evaluate the performance across different feature sets.

2.7 Full *E. coli* AST GVR Dataset

This process was repeated for the full *E. coli* GVR dataset, covering bacterial growth under different antibiotics. Each part was completed in a specific Jupyter Notebook, which was duplicated and adapted to the full dataset. The preprocessing and feature engineering pipeline was adapted to accommodate the dataset. The *E. coli* dataset included 5 antibiotics (ampicillin, ciprofloxacin, gentamicin, tetracycline, and trimethoprim) across all available experiments, excluding the previously identified ampicillin outlier. The dataset was composed of 45342 rows, with the labels' growth (1) 27768 and growth (0) 17574. After test/ train splitting, the train dataset had 42372 rows, and the test 1 (ampicillin_20230518_5e4) and test 2 (ampicillin_20230518_5e6) had 990 rows each. Model evaluation was conducted using the same experimental splits and feature combinations to assess consistency and evolution.

2.8 Cross-antibiotic Evaluation

To assess generalisation across antibiotics, the model was further evaluated on experiments for each of the remaining antibiotics: ampicillin_20230518_5e4, ciprofloxacin_20250719_5e5, gentamicin_20250728_5e5, tetracycline_20240203_5e5, and trimethoprim_20250717_5e5. All were excluded from the training set. Accuracy-over-time plots were generated to evaluate whether the model trained on the full *E. coli* data could maintain predictive performance across antibiotics and experimental conditions.

2.9 Software and Code Availability

This project was implemented using Python 3.12.6 and developed in Visual Studio Code. The following Python libraries were used: NumPy, pandas, scikit-learn, and Matplotlib. [28]–[30] The code used in this project is available at <https://github.com/FiSantos/project-ml-for-ast>.

3 Results and Discussion

3.1 Model Selection

Several classification algorithms were evaluated using accuracy, precision, recall, F1-score, ROC-AUC, and confusion matrices across both datasets (Table 1). As models were trained on the complete time series, the reported metrics represent conservative performance estimates.

Naïve Bayes was excluded due to anomalous behaviour characterised by a high false negative rate, in which growing bacteria were systematically misclassified as non-growing. This is particularly problematic in AST, where false negatives could lead to inappropriate treatment decisions. This failure is consistent with the known limitation of the conditional independence assumption underlying Naïve Bayes, which is not present here, as most features derive from the same growth signal and are therefore highly correlated. [31], [32]

Among the remaining classifiers, Random Forest consistently achieved the highest accuracy and ROC-AUC while maintaining the most balanced confusion matrices, 0.76 and 0.87 in the *E. coli*–ampicillin dataset, and 0.75 and 0.88 in the full *E. coli* dataset. The stability of these results across datasets suggests robust generalisation to different growth conditions. Although SVM showed competitive performance, its training time was substantially longer (3 min 21.2 s versus 7.6 s for Random Forest), which is a relevant consideration in time-sensitive clinical workflows. Random Forest was

therefore selected as the final model, consistent with previous studies reporting its strong performance in biological classification tasks. [33]

Table 1. Model performance metrics for (A) The *E. coli*-ampicillin microfluidic AST, based on the GVR dataset (Test1:20230518_5e4). (B) The full *E. coli* microfluidic AST based on the GVR dataset (Test 1: 20230518_5e4)

A	Metrics Models	Accuracy	F1-Score		Recall		Precision		AUC-ROC	Cross Validation Mean	Confusion Matrix
			0	1	0	1	0	1			
	Logistic Regression	0.75	0.72	0.78	0.63	0.87	0.83	0.70	0.84	0.90	[[314 181] [66 429]]
	Random Forest	0.76	0.76	0.78	0.74	0.80	0.79	0.75	0.87	0.91	[[365 130] [98 397]]
	K-Nearest Neighbours	0.72	0.70	0.69	0.63	0.83	0.79	0.69	0.78	0.87	[[312 183] [85 410]]
	Support Vector Machine	0.74	0.71	0.77	0.62	0.87	0.83	0.70	0.73	0.88	[[305 190] [62 433]]
	Naive Bayes	0.75	0.80	0.67	1.00	0.51	0.67	1.00	0.80	0.83	[[495 0] [244 251]]
B	Metrics Models	Accuracy	F1-Score		Recall		Precision		AUC-ROC	Cross Validation Mean	Confusion Matrix
			0	1	0	1	0	1			
	Logistic Regression	0.73	0.72	0.75	0.68	0.80	0.77	0.71	0.86	0.90	[[337 158] [100 395]]
	Random Forest	0.75	0.79	0.69	0.96	0.55	0.68	0.93	0.88	0.94	[[473 22] [223 272]]
	K-Nearest Neighbours	0.73	0.73	0.74	0.73	0.74	0.74	0.73	0.82	0.91	[[359 136] [128 367]]
	Support Vector Machine	0.75	0.74	0.78	0.67	0.85	0.81	0.72	0.83	0.93	[[332 163] [76 419]]
	Naive Bayes	0.74	0.79	0.49	0.99	0.49	0.66	0.98	0.84	0.85	[[491 4] [250 245]]

3.2 Feature Importance

Feature importance was assessed to determine whether the model was learning biologically meaningful patterns or exploiting shortcuts in the data, as shown in Table 2. This is a critical question given that the model had a perfect score on the *E. coli*-ampicillin dataset when antibiotic concentration was included.

Table 2. Feature importance for both datasets used with and without concentration

Feature		Concentration	AUC	Slope	GVR	Delta	Current_STD	Tube_STD	Time	Initial_STD	Derivative
Importance	Ecoli - Ampicillin:	0.50	0.10	0.10	0.08	0.05	0.05	0.03	0.02	0.01	0.005
	Ecoli - Ampicillin: No		0.22	0.12	0.19	0.11	0.12	0.07	0.07	0.02	0.03
	Ecoli: Concentration	0.15	0.17	0.09	0.24	0.13	0.05	0.04	0.02	0.05	0.007
	Ecoli: No Concentration		0.22	0.06	0.35	0.11	0.06	0.06	0.04	0.05	0.02

In the *E. coli*-ampicillin dataset, concentration dominated with an importance of 0.50. With only ampicillin experiments and a fixed MIC between two concentration points, the model could memorise the growth-inhibition boundary without learning growth dynamics, a case of hidden data leakage. In the full *E. coli* dataset, concentration

importance dropped to 0.15, confirming this shortcut was specific to the single-antibiotic structure. Concentration was then excluded from subsequent modelling, as a clinically deployable model must base predictions on observed bacterial behaviour rather than antibiotic concentration. After its removal, AUC, GVR, Slope, Delta, and Current Standard Deviation emerged as dominant predictors, all reflecting bacterial growth rate and quantity. However, as AUC was also used as the sole criterion for defining growth labels, its high importance may partly reflect circularity rather than independent predictive power, a limitation that future work should address through alternative labelling strategies. Furthermore, as the analysis is restricted to *E. coli* and five antibiotics, whether the same features dominate in other species or with different mechanisms of action remains unknown.

3.3 Model Testing

To evaluate model performance over time, accuracy was monitored at each time point across both test sets and datasets, with the most representative results shown in Figure 3. Prediction time was defined as the earliest point at which accuracy reached 0.90 and was not reduced for the remainder of the experiment, a criterion chosen to ensure clinical reliability, as unstable predictions were considered unfit regardless of great performance.

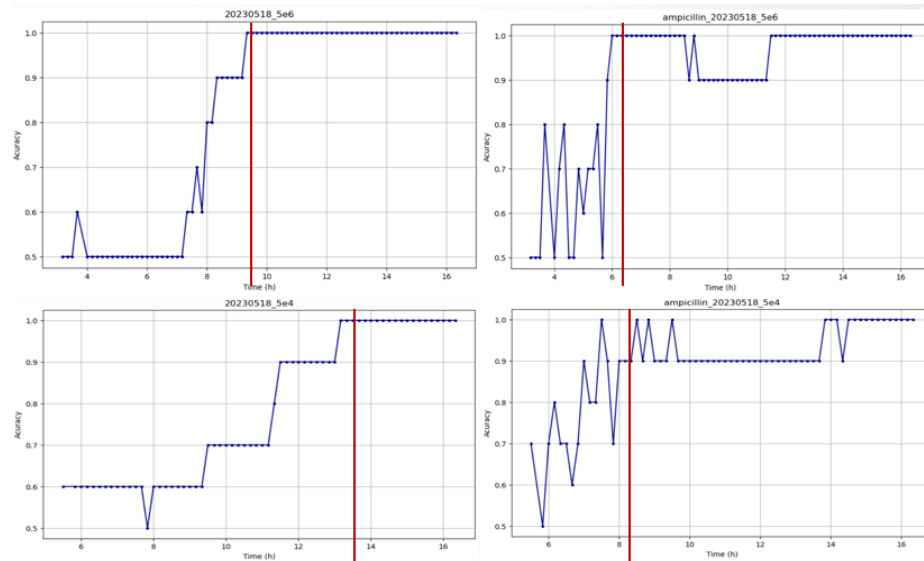


Fig 3. Accuracy per time for Test 1(5×10^6 CFU/ml, top) and Test 2(5×10^4 CFU/ml, bottom), trained on the *E. coli*-ampicillin dataset (left) and the full *E. coli* dataset (right). The red vertical line indicates the time point where accuracy exceeds 0.9.

Test Sets 1 and 2 differ in bacterial concentration (5×10^6 and 5×10^4 CFU/mL, respectively), which directly influenced prediction timing. At higher inoculum levels, GVR divergence appeared earlier, and 0.90 accuracy was reached at 8.33 h, reducing to 5.83 h with the full *E. coli* model. At lower concentration, prediction time decreased from 11.49 h to 8 h. This is consistent with bacterial growth kinetics, where a larger inoculum reaches detectable population thresholds sooner, and suggests the model responds to genuine biological growth signals rather than experimental artefacts.

The most significant finding was the reduction in prediction time from approximately 12 h to 8 h after expanding the training dataset. This improvement reflects the greater diversity of growth trajectories in the full dataset. Exposure to multiple antibiotics with distinct mechanisms of action allowed the model to learn more generalised early-stage growth patterns. This suggests that dataset diversity is important for model performance. While 6–8 h is longer than some rapid phenotypic methods, such as deep learning-based video microscopy, attaining MIC inference within 30 minutes. It represents a substantial improvement over conventional AST, which typically requires 16–24 h. In a clinical context, earlier susceptibility estimation could support more targeted antibiotic therapy, reduce broad-spectrum antibiotic use, and ultimately contribute to combating antimicrobial resistance. [1], [3], [25]

3.4 Cross-antibiotic Evaluation

To assess model generalisation, performance was evaluated independently for each antibiotic, as shown in Figure 4. Prediction time was approximately 8h for ampicillin, tetracycline, and gentamicin, and approximately 5h for ciprofloxacin and trimethoprim. As ampicillin experiments used a lower bacterial concentration (5×10^4 vs 5×10^5 CFU/mL), direct comparison with the remaining antibiotics is limited. However, as lower concentrations are associated with slower growth detection, the 8h result likely represents a conservative estimate.

The shorter prediction times for ciprofloxacin and trimethoprim may reflect more clearly distinguishable early growth patterns associated with their mechanisms of action. However, as only two training experiments were available for each, further data are required before drawing definitive conclusions. The cross-antibiotic evaluation was conducted on a model trained with a limited number of experiments overall, which may affect the robustness of the observed performance. Altogether, these findings suggest that a Random Forest model trained on growth dynamics features can reliably predict antimicrobial susceptibility in *E. coli* within 8h across multiple antibiotics, without relying on experimental metadata. While further validation with additional species, antibiotics, and larger datasets is required, these results represent a promising step towards rapid AST.

Compared with existing ML-based rapid AST approaches, such as CNN-based video *Yu et al. (2018)*, the present model operates on a longer timescale. [25] However, it successfully reduced prediction time from the conventional mcAST endpoint of 12 h to 8 h, with 10 h as a more conservative threshold. This is a reduction of 4-2h that in a clinical setting could meaningfully accelerate treatment decisions. These results demonstrate that simple microfluidic growth dynamics carry sufficient information for ML-based susceptibility prediction.

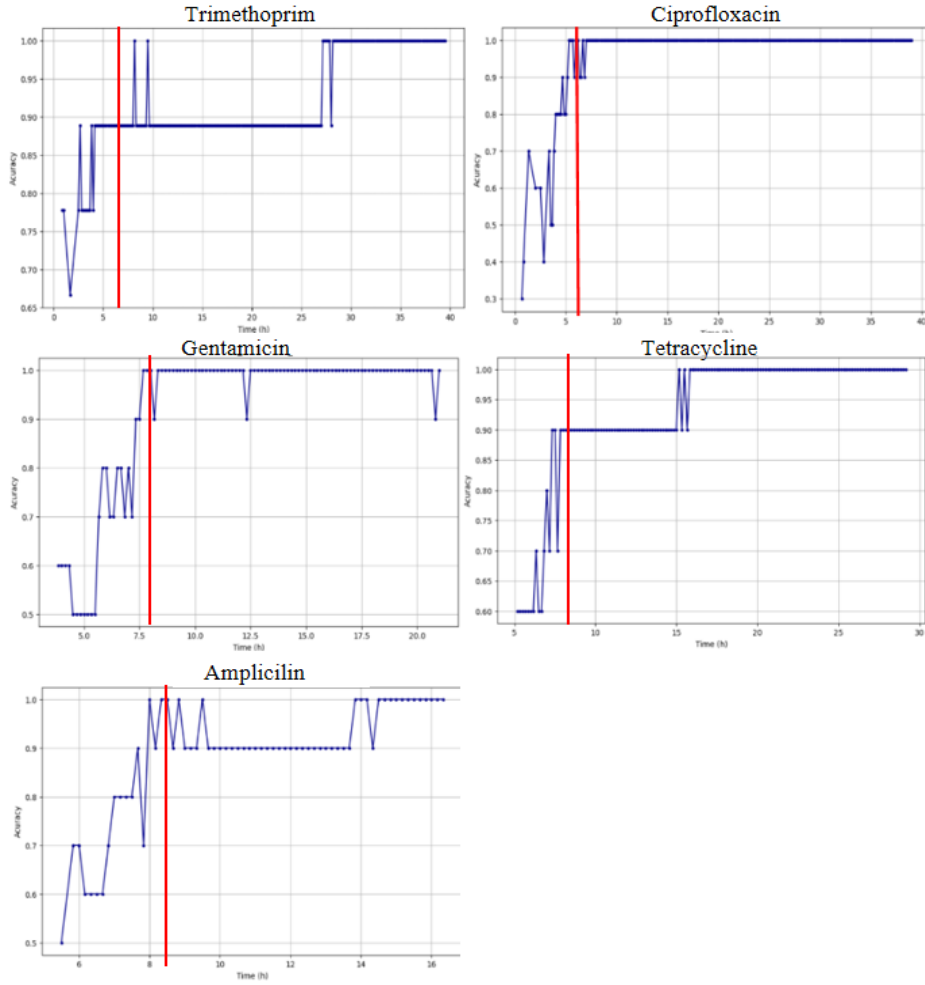


Fig 4. Accuracy per time for trimethoprim(5×10^5), ciprofloxacin(5×10^5), gentamicin(5×10^5), tetracycline (5×10^5) and ampicillin(5×10^4) CFU/ml, trained on the full *E. coli* dataset. The red vertical line indicates the time point where accuracy exceeds 0.9.

4 Conclusion

This project aimed to assess whether ML models can use early GVR time-series data to predict bacterial growth and enable earlier antimicrobial susceptibility estimation. The results demonstrate that this is possible. The model successfully learned from bacterial growth kinetics, reducing prediction time from the conventional mcAST endpoint of 12 h to 8 h, a reduction of 4 h. This represents a considerable improvement over conventional AST, which typically requires 16–24 h. Feature importance analysis suggests that the model learns from biologically meaningful growth dynamics rather than experimental metadata, supporting its potential for clinical applicability.

Future work should integrate *S. aureus* data, evaluate performance on clinical samples, and develop a robust, automated pipeline capable of processing GVR data in real time, enabling direct integration into clinical workflows where susceptibility results would be generated automatically as the mcAST experiment runs.

References

- [1] C. S. Ho *et al.*, ‘Antimicrobial resistance: a concise update’, *The Lancet Microbe*, vol. 6, no. 1, pp. 1–14, 2025, doi: 10.1016/j.lanmic.2024.07.010.
- [2] ‘Antimicrobial resistance’, world health organization. Accessed: Feb. 28, 2026. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/antimicrobial-resistance>
- [3] I. Gajic *et al.*, ‘Antimicrobial Susceptibility Testing: A Comprehensive Review of Currently Used Methods’, *Antibiotics*, vol. 11, no. 4, pp. 1–26, 2022, doi: 10.3390/antibiotics11040427.
- [4] M. A. Salam, M. Y. Al-Amin, J. S. Pawar, N. Akhter, and I. B. Lucy, ‘Conventional methods and future trends in antimicrobial susceptibility testing’, *Saudi J. Biol. Sci.*, vol. 30, no. 3, p. 103582, 2023, doi: 10.1016/j.sjbs.2023.103582.
- [5] C. Li *et al.*, ‘Portable intelligent digital microfluidic system for rapid antibiotic susceptibility testing’, *Chem. Eng. J.*, vol. 512, p. 162205, May 2025, doi: 10.1016/j.cej.2025.162205.
- [6] J. Xu *et al.*, ‘Optical phenotypic tracking: Advancing rapid toward culture-free antimicrobial susceptibility testing’, *TrAC Trends Anal. Chem.*, vol. 197, p. 118700, Apr. 2026, doi: 10.1016/j.trac.2026.118700.
- [7] P. Bharathiraja and E. Tamilmani, ‘Roadmap for emerging antimicrobial susceptibility testing and its clinical adaptation, from bench to bedside’, *J. Microbiol. Methods*, vol. 241, p. 107386, Feb. 2026, doi: 10.1016/j.mimet.2025.107386.
- [8] S. Gopalakrishnan, D. Mall, S. Pushpavanam, and R. Karmakar, ‘Rapid antimicrobial susceptibility testing using carbon screen printed electrodes in a microfluidic device’, *Sci. Rep.*, vol. 15, no. 1, pp. 1–12, 2025, doi: 10.1038/s41598-024-84286-3.
- [9] S. Meng *et al.*, ‘Rapid antimicrobial susceptibility testing (AST) based on in-situ time-lapse imaging of microcolonies’, *Sensors Actuators B Chem.*, vol. 448, p. 139012, Feb.

- 2026, doi: 10.1016/J.SNB.2025.139012.
- [10] G. Resznetnik *et al.*, ‘Next-generation rapid phenotypic antimicrobial susceptibility testing’, *Nat. Commun.*, vol. 15, no. 1, 2024, doi: 10.1038/s41467-024-53930-x.
 - [11] K. P. Smith and J. E. Kirby, ‘Rapid Susceptibility Testing Methods’, *Clin. Lab. Med.*, vol. 39, no. 3, pp. 333–344, 2019, doi: 10.1016/j.cll.2019.04.001.
 - [12] S. İ. Dönmez, S. H. Needs, H. M. I. Osborn, N. M. Reis, and A. D. Edwards, ‘Label-free 1D microfluidic dipstick counting of microbial colonies and bacteriophage plaques’, *Lab Chip*, vol. 22, no. 15, pp. 2820–2831, 2022, doi: 10.1039/d2lc00280a.
 - [13] S. İ. Dönmez, S. H. Needs, H. M. I. Osborn, and A. D. Edwards, ‘Label-free smartphone quantitation of bacteria by darkfield imaging of light scattering in fluoropolymer micro capillary film allows portable detection of bacteriophage lysis’, *Sensors Actuators, B Chem.*, vol. 323, no. March, 2020, doi: 10.1016/j.snb.2020.128645.
 - [14] S. H. Needs *et al.*, ‘Moving microcapillary antibiotic susceptibility testing (mcAST) towards the clinic: unravelling kinetics of detection of uropathogenic *E. coli*, mass-manufacturing and usability for detection of urinary tract infections in human urine†’, *Sensors and Diagnostics*, vol. 2, no. 3, pp. 736–750, 2023, doi: 10.1039/d2sd00138a.
 - [15] N. M. Reis, J. Pivetal, A. L. Loo-Zazueta, J. M. S. Barros, and A. D. Edwards, ‘Lab on a stick: Multi-analyte cellular assays in a microfluidic dipstick’, *Lab Chip*, vol. 16, no. 15, pp. 2891–2899, 2016, doi: 10.1039/c6lc00332j.
 - [16] and H. Y. Bing Li, Yong Qiu, Andrew Glidle, David McIlvenna, Qian Luo, Jon Cooper, Han-Chang Shi, ‘Gradient Microfluidics Enables Rapid Bacterial Growth Inhibition Testing’, *Anal. Chem.*, vol. 86, no. 6, pp. 3131–3137, 2014, doi: DOI: 10.1021/ac5001306.
 - [17] C. G. Giske *et al.*, ‘Update from the European Committee on Antimicrobial Susceptibility Testing (EUCAST)’, *J. Clin. Microbiol.*, vol. 60, no. 3, pp. 1–10, 2022, doi: 10.1128/jcm.00276-21.
 - [18] M. S. Badar, ‘A Guide to Applied Machine Learning for Biologists’, *A Guid. to Appl. Mach. Learn. Biol.*, vol. 23, no. January, p. C1, 2023, doi: 10.1007/978-3-031-22206-1_11.
 - [19] A. Smiti, ‘When machine learning meets medical world: Current status and future challenges’, *Comput. Sci. Rev.*, vol. 37, p. 100280, Aug. 2020, doi: 10.1016/j.cosrev.2020.100280.
 - [20] S. Valavarasu, Y. Sangu, and T. Mahapatra, ‘Prediction of antibiotic resistance from antibiotic susceptibility testing results from surveillance data using machine learning’, *Sci. Rep.*, vol. 15, no. 1, pp. 1–12, 2025, doi: 10.1038/s41598-025-14078-w.
 - [21] M. Mediouni, V. Makarenkov, and A. B. Diallo, ‘Towards an interpretable machine learning model for predicting antimicrobial resistance’, *J. Glob. Antimicrob. Resist.*, vol. 45, pp. 47–51, 2025, doi: 10.1016/j.jgar.2025.08.011.
 - [22] M. Yasir, A. M. Karim, S. K. Malik, A. A. Bajaffer, and E. I. Azhar, ‘Prediction of antimicrobial minimal inhibitory concentrations for *Neisseria gonorrhoeae* using machine learning models’, *Saudi J. Biol. Sci.*, vol. 29, no. 5, pp. 3687–3693, 2022, doi: 10.1016/j.sjbs.2022.02.047.
 - [23] J. C. Osval Antonio Montesinos López, Abelardo Montesinos López, *Multivariate Statistical Machine Learning Methods for Genomic Prediction*. Springer Cham, 2022. doi: <https://doi.org/10.1007/978-3-030-89010-0>.

- [24] M. Salvi, U. R. Acharya, F. Molinari, and K. M. Meiburger, ‘The impact of pre- and post-image processing techniques on deep learning frameworks: A comprehensive review for digital pathology image analysis’, *Comput. Biol. Med.*, vol. 128, p. 104129, Jan. 2021, doi: 10.1016/j.combiomed.2020.104129.
- [25] H. Yu *et al.*, ‘Phenotypic Antimicrobial Susceptibility Testing with Deep Learning Video Microscopy’, *Anal. Chem.*, vol. 90, no. 10, pp. 6314–6322, May 2018, doi: 10.1021/acs.analchem.8b01128.
- [26] G. B. Biffignandi, L. Chindelevitch, M. Corbella, E. J. Feil, and D. Sasser, ‘Optimising machine learning prediction of minimum inhibitory concentrations in *Klebsiella pneumoniae*’, pp. 1–15, 2024, doi: 10.1099/mgen.0.001222.
- [27] A. Sturm *et al.*, ‘Accurate and rapid antibiotic susceptibility testing using a machine learning-assisted nanomotion technology platform’, 2024, doi: 10.1038/s41467-024-46213-y.
- [28] T. pandas development Team, ‘pandas-dev/pandas: Pandas’. Zenodo, 2020. doi: 10.5281/zenodo.3509134.
- [29] C. R. Harris *et al.*, ‘Array programming with NumPy’, *Nature*, vol. 585, no. 7825, pp. 357–362, 2020, doi: 10.1038/s41586-020-2649-2.
- [30] J. D. Hunter, ‘Matplotlib: A 2D graphics environment’, *Matplotlib A 2D Graph. Environ.*, vol. 9, pp. 90–95, 2007, doi: 10.1109/MCSE.2007.55.
- [31] W. Wang, L. Yan, F. Liu, and Y. Li, ‘Improving Gaussian Naive Bayes classification on imbalanced data through coordinate-based minority feature mining.’, *PeerJ. Comput. Sci.*, vol. 11, p. e3003, 2025, doi: 10.7717/peerj-cs.3003.
- [32] T. Kim and J.-S. Lee, ‘Maximizing AUC to learn weighted naive Bayes for imbalanced data classification’, *Expert Syst. Appl.*, vol. 217, p. 119564, 2023, doi: <https://doi.org/10.1016/j.eswa.2023.119564>.
- [33] R. Liu, B. Yang, E. Zio, and X. Chen, ‘Artificial intelligence for fault diagnosis of rotating machinery: A review’, *Mech. Syst. Signal Process.*, vol. 108, pp. 33–47, 2018, doi: <https://doi.org/10.1016/j.ymssp.2018.02.016>.

Acknowledgments

The authors would like to thank Zhuoling Yu for providing the experimental dataset and for her valuable input and discussions, and Alexandre Pascoal and Fabiana Goes for their helpful feedback and suggestions.